

Conjecture 00000002511: symmetric tensors are homogeneous polynomials, and the dictionary is multiplicative

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1 Statement and reading

Conjecture (verbatim). *Definition: Symmetric tensors: the symmetric version. Conjecture: The symmetric version of tensors corresponds to homogeneous polynomials: symmetric tensors are forms, and the dictionary closes under homogeneous polynomials.*

Reading. This is the classical dictionary between symmetric tensors and homogeneous polynomials (forms); we prove it as a known result, it is not new. We fix a field K of characteristic 0, a K -vector space V with a finite basis $b = (b_j)_{j \in \iota}$ ($n = |\iota|$), and a degree $d \geq 0$. We prove:

- (a) *Symmetric tensors are forms:* the dictionary map sends every symmetric d -tensor to a homogeneous polynomial of degree d in the n variables X_j .
- (b) *Correspondence:* restricted to symmetric tensors, the dictionary is a linear bijection $\text{Sym}^d(V) \rightarrow K[X_j : j \in \iota]_d$.
- (c) *The dictionary closes under products of forms:* the symmetric product $T \odot U$ of symmetric tensors is sent to the product of the two forms, and conversely the product pq of two forms corresponds to the symmetric product of the tensors of p and q .

Conventions. (i) Characteristic 0 is assumed (the standard setting; both sources below work in it). We do not consider other characteristics. (ii) $d = 0$ is allowed: $V^{\otimes 0} = K$ and the forms of degree 0 are the constants. (iii) Product convention: the symmetrisation is the *averaging* operator $\text{Sym } t = \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot t$ and $T \odot U = \text{Sym}(T \otimes U)$, where $T \otimes U$ puts the d factors of T before the e factors of U . Only this normalisation is formalised. (iv) The polynomials are formal polynomials in the coordinates X_j ; Theorem 3 identifies them with the polynomial functions $\varphi \mapsto T(\varphi, \dots, \varphi)$ on V^* .

2 Definitions (as formalised)

Definition 1. $V^{\otimes d}$ is Mathlib's `PiTensorProduct` $\bigotimes_{k \in \text{Fin } d} V$ (`TPow K V d`). A permutation $\sigma \in S_d$ acts by permuting the tensor factors, $\sigma \cdot (v_1 \otimes \dots \otimes v_d) = v_{\sigma^{-1}(1)} \otimes \dots \otimes v_{\sigma^{-1}(d)}$ (`permAct`, which is `PiTensorProduct.reindex`). The *symmetric tensors* $\text{Sym}^d(V)$ (`symTensors K V d`) are the tensors fixed by every $\sigma \in S_d$.

This is the definition in the retrieved Wikipedia article: “Then T is a symmetric tensor if $[\tau_\sigma T = T]$ for the braiding maps associated to every permutation σ on the symbols $\{1, 2, \dots, k\}$ ”.

Definition 2. The generic linear form is $\ell(v) = \sum_j b_j^*(v) X_j$, i.e. the linear map with $b_j \mapsto X_j$ (`linForm`). The dictionary map $P = P_{b,d} : V^{\otimes d} \rightarrow K[X_j : j \in \iota]$ (`toPoly`) is the linear map with $P(v_1 \otimes \cdots \otimes v_d) = \ell(v_1) \cdots \ell(v_d)$ (defined by the universal property `PiTensorProduct.lift`). Forms of degree d are Mathlib’s `MvPolynomial.homogeneousSubmodule` ι K d .

Theorem 3 (`eval_toPoly`). For every linear functional $\varphi \in V^*$ and $T \in V^{\otimes d}$, $P(T)((\varphi(b_j))_j) = (\varphi \otimes \cdots \otimes \varphi)(T)$, i.e. $P(T)$ is the polynomial $\varphi \mapsto T(\varphi, \dots, \varphi)$ written in the coordinates of φ .

Proof. Both sides are linear in T ; on $v_1 \otimes \cdots \otimes v_d$ both equal $\prod_k \varphi(v_k)$, since evaluation at $(\varphi(b_j))_j$ composed with ℓ is φ (check on the basis). $\square$

3 Proof

Let $B_i = b_{i_1} \otimes \cdots \otimes b_{i_d}$, $i : \text{Find} \rightarrow \iota$, be the product basis of $V^{\otimes d}$ (`tBasis`, Mathlib’s `Basis.piTensorProduct`) and T_i the coordinates of T . The *content* of a word i is the exponent vector $c(i) = \sum_k e_{i_k} \in \mathbb{N}^\iota$ (`content`); $c(i)_a = \#\{k : i_k = a\}$ and $|c(i)| = d$.

Lemma 4. (1) $P(B_i) = X^{c(i)}$, hence $P(T) = \sum_i T_i X^{c(i)}$ and the coefficient of X^α in $P(T)$ is $\sum_{c(i)=\alpha} T_i$ (`toPoly_tBasis`, `toPoly_eq_sum`, `coeff_toPoly`). (2) $(\sigma \cdot T)_i = T_{i \circ \sigma}$ (`repr_permAct`). (3) If $c(i) = c(j)$ then $i = j \circ \sigma$ for some $\sigma \in S_d$ (`exists_perm_of_content_eq`: the fibres $\{k : i_k = a\}$ and $\{k : j_k = a\}$ have equal size, and `Equiv.ofFiberEquiv` glues bijections between them). Hence a symmetric T has $T_i = T_j$ whenever $c(i) = c(j)$ (`repr_sym_eq`). (4) Every $\alpha \in \mathbb{N}^\iota$ with $|\alpha| = d$ is $c(i)$ for some i (`exists_content_eq`, induction on d , removing one unit from a nonzero entry).

(a) *Forms.* By Lemma 4(1), $P(T)$ is a combination of monomials of degree d , so $P(T)$ is homogeneous of degree d for every T , in particular for symmetric T (`toPoly_mem`).

(b) *Injectivity.* Let T be symmetric with $P(T) = 0$ and fix i . With $F = \{j : c(j) = c(i)\}$, Lemma 4(1),(3) give $0 = \sum_{j \in F} T_j = |F| T_i$. Since $i \in F$, $|F| \geq 1$, and $|F| \neq 0$ in K because $\text{char } K = 0$; so $T_i = 0$ (`toPoly_eq_zero`).

(b) *Surjectivity.* Let $|\alpha| = d$, $F_\alpha = \{j : c(j) = \alpha\}$ (nonempty by Lemma 4(4)) and $S_\alpha = |F_\alpha|^{-1} \sum_{j \in F_\alpha} B_j$. By Lemma 4(2) and $c(j \circ \sigma) = c(j)$, S_α is symmetric, and by (1) $P(S_\alpha) = X^\alpha$ (`exists_sym_monomial`). The forms of degree d are spanned by the monomials X^α with $|\alpha| = d$ (Mathlib `homogeneousSubmodule_eq_finsupp_supported`), so every form of degree d is P of a symmetric tensor (`exists_sym_of_mem`). Hence the restriction `symToForm` b d is bijective (`symToForm_bijective`) and gives the linear isomorphism `symEquiv` b d : $\text{Sym}^d(V) \simeq K[X]_d$; in particular the two spaces have equal dimension (`finrank_symTensors`).

(c) *Products.* Let $T \otimes U \in V^{\otimes(d+e)}$ be `reindex` `finSumFinEquiv` (`tmulEquiv` ($T \otimes U$)) (`tmulL`). On pure tensors $P(T \otimes U) = \prod_{k < d+e} \ell(\cdot) = P(T)P(U)$ (split the product with `Fin.prod_univ_add`), so by bilinearity $P(T \otimes U) = P(T)P(U)$ (`toPoly_tmulL`). Also $P(\sigma \cdot t) = P(t)$ (the product of the $\ell(v_k)$ is reordered, `toPoly_permAct`), hence $P(\text{Sym } t) = \frac{1}{d!} \cdot d! \cdot P(t) = P(t)$ (`toPoly_symz`). We check that `Sym` t is symmetric (`symz_mem`) and that `Sym` fixes symmetric tensors (`symz_of_mem`). Therefore $P(T \odot U) = P(T \otimes U) = P(T)P(U)$ (`symToForm_symProd`). Applying the inverse isomorphisms gives the converse statement in (c).

4 Lean formalisation

File `Conjecture2511/Basic.lean` (namespace `C2511`, 341 lines, only `import Mathlib`). The main theorem is

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theorem symmetric_tensors_are_forms [CharZero K] (b : Module.Basis iota K V) :
  (forall (d : Nat) (T : TPow K V d), T mem symTensors K V d ->
    toPoly b d T mem homogeneousSubmodule iota K d) /\
  (forall d : Nat, Function.Bijective (symToForm b d)) /\
  (forall (d e : Nat) (T : symTensors K V d) (U : symTensors K V e),
    (symToForm b (d + e) (symProd T U) : MvPolynomial iota K) =
      symToForm b d T * symToForm b e U) /\
  (forall (d e : Nat) (p : homogeneousSubmodule iota K d)
    (q : homogeneousSubmodule iota K e),
    (symEquiv b (d + e)).symm <p * q, p.2.mul q.2> =
      symProd ((symEquiv b d).symm p) ((symEquiv b e).symm q))

```

(ASCII transliteration of the Lean source; ι is an arbitrary Fintype, V any K -vector space with basis b). Supporting results: `eval_toPoly`, `symEquiv`, `finrank_symTensors`, `symz_of_mem`. No component from earlier packages was reused.

Axiom audit. `lake build` succeeds; `#print axioms` for `symmetric_tensors_are_forms`, `eval_toPoly`, `symEquiv`, `finrank_symTensors` and `symz_of_mem` each reports `[proptext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

5 Sources (retrieved)

- Wikipedia, *Symmetric tensor*, https://en.wikipedia.org/wiki/Symmetric_tensor: “The space of symmetric tensors of order r on a finite-dimensional vector space V is naturally isomorphic to the dual of the space of homogeneous polynomials of degree r on V . Over fields of characteristic zero, the graded vector space of all symmetric tensors can be naturally identified with the symmetric algebra on V .” The same article defines the averaging symmetrisation and the product: “Given two tensors $T_1 \in \text{Sym}^k_1(V)$ and $T_2 \in \text{Sym}^k_2(V)$, we use the symmetrization operator to define:” $T_1 \odot T_2 = \text{Sym}(T_1 \otimes T_2)$.
- P. Comon, G. Golub, L.-H. Lim, B. Mourrain, *Symmetric tensors and symmetric tensor rank*, SIAM J. Matrix Anal. Appl. (2008), <https://arxiv.org/abs/0802.1681>, Section 3.1 “Equivalence with homogeneous polynomials”: “every symmetric tensor of order and dimension may be uniquely associated with a homogeneous polynomial of degree in variables” and “the correspondence between symmetric tensors and homogeneous polynomials is obviously bijective” (mathematical symbols were lost in the retrieved text).